
Algorithm 1 World Action Planner (WAP)

Require: VLM Agent, robot controller ϕ , world model WM, optional policy π

Input: Task description ℓ , initial state s , environment env

while not done **do**

▷ **Agent Action Proposal**

Propose primitive action goals using the VLM agent: $g = \text{Agent}.\text{Proposal}(s, \ell)$

Translate the proposed goal into robot actions: $a = \phi(g, s_{\text{proprio}})$

▷ **Global Optimization**

Imagine the proposed action using the world model: $\hat{s}_{\text{next}} = \text{WM}(s, a)$

Obtain agent feedback: $\Delta g = \text{Agent}.\text{Eval}(\hat{s}_{\text{next}}, \ell)$

Update the goal and robot action: $g \leftarrow g + \Delta g, a = \phi(g, s_{\text{proprio}})$

▷ **Local Search and Refinement**

Generate local goal candidates: $g_1, g_2, \dots, g_N = \text{GridSearch}(g)$

Translate goal candidates into robot actions: $a_i = \phi(g_i, s_{\text{proprio}}), i = 1, 2, \dots, N$

Imagine the candidate actions: $\hat{s}_i = \text{WM}(s, a_i), i = 1, 2, \dots, N$

if use policy **then**

Imagine the policy rollout after each candidate action: $\hat{s}_i = \text{WM}(\hat{s}_i, \pi(\hat{s}_i)), i = 1, 2, \dots, N$

end if

Select the best candidate using the VLM agent: $i^* = \text{Agent}.\text{Rank}(\hat{s}_1, \hat{s}_2, \dots, \hat{s}_N, \ell)$

▷ **Execute the Actions**

Execute the selected action: $s, \text{done} = \text{env}.\text{Step}(s, a_{i^*})$

if use policy **then**

Roll out the policy following the selected action: $s, \text{done} = \text{env}.\text{Step}(s, \pi(s))$

end if

end while
